Google — 30 DSA Interview Practice Questions

Targeted practice set for software-engineering interview preparation. Problems are LeetCode-style and can be practiced on LeetCode, GeeksforGeeks, HackerRank, CodeSignal, CodeChef or equivalent platforms. This is a preparation list based on public interview patterns; it is not a claim that these exact questions will be asked.

Public reports show LeetCode-style DSA/graph-reachability problem solving in recent interviews.

#	Problem	Topic	Difficulty	Platform
1	Best Time to Buy and Sell Stock	Array	Easy	LeetCode-style
2	Binary Search	Binary Search	Easy	LeetCode-style
3	Climbing Stairs	DP	Easy	LeetCode-style
4	Diameter of Binary Tree	Tree/DFS	Easy	LeetCode-style
5	Maximum Depth of Binary Tree	Tree	Easy	LeetCode-style
6	Two Sum	Array/Hashing	Easy	LeetCode-style
7	Valid Parentheses	Stack	Easy	LeetCode-style
8	0/1 Knapsack	DP	Medium	LeetCode-style
9	Binary Tree Level Order Traversal	Tree/BFS	Medium	LeetCode-style
10	Cheapest Flights Within K Stops	Graph/DP	Medium	LeetCode-style
11	Coin Change	DP	Medium	LeetCode-style
12	Decode Ways	DP	Medium	LeetCode-style
13	Find Minimum in Rotated Sorted Array	Binary Search	Medium	LeetCode-style
14	House Robber	DP	Medium	LeetCode-style
15	Koko Eating Bananas	Binary Search on Answer	Medium	LeetCode-style
16	Longest Common Subsequence	DP	Medium	LeetCode-style
17	Longest Increasing Subsequence	DP	Medium	LeetCode-style
18	Longest Substring Without Repeating Characters	Sliding Window	Medium	LeetCode-style
19	Lowest Common Ancestor of a Binary Tree	Tree	Medium	LeetCode-style
20	Maximal Square	DP/Matrix	Medium	LeetCode-style
21	Number of Islands	Graph/BFS-DFS	Medium	LeetCode-style
22	Search in Rotated Sorted Array	Binary Search	Medium	LeetCode-style
23	Ugly Number II	DP/Heap	Medium	LeetCode-style
24	Word Break	DP	Medium	LeetCode-style
25	Binary Tree Maximum Path Sum	Tree/DFS	Hard	LeetCode-style
26	Distinct Subsequences	DP	Hard	LeetCode-style
27	Edit Distance	DP	Hard	LeetCode-style
28	Median of Two Sorted Arrays	Binary Search	Hard	LeetCode-style
29	Minimum Window Substring	Sliding Window	Hard	LeetCode-style
30	Serialize and Deserialize Binary Tree	Tree/Design	Hard	LeetCode-style

Interview practice rule: For each problem, first explain brute force, then optimize it. Always state time complexity, space complexity, assumptions and edge cases. After solving, practice one variation of the same pattern.